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Module 7: Assignment
Information disclosure and application architecture
MCY 670 Web Security

Questions

1. message:

(15 points) While attempting to launch malformed requests to the web server, an attacker encounters the following error

```
Warning: mysql_connect() [function.mysql-connect]: Access denied
for user 'premiumdde'@'localhost' (using password: YES) in
/home/doau/public_html/premiumdde/directory on line 15
Warning: mysql_select_db() [function.mysql-select-db]: Access
denied for user 'nobody'@'localhost' (using password: NO) in
/home/doau/public_html/premiumdde/directory on line 16
Warning: mysql_select_db() [function.mysql-select-db]: A link to
the server could not be established in
/home/doau/public_html/premiumdde/directory on line 16
Warning: mysql_query() [function.mysql-query]: Access denied for
user 'nobody'@'localhost' (using password: NO) in
/home/doau/public_html/premiumdde/directory on line 448
```

What useful pieces of information can the attacker extract from this error message?

The real MySQL usernames being used (premiumdde and nobody), the fact that the application uses out of date PHP MySQL functions, and the possibility that one of the accounts doesn't need a password are just a few of the important facts that the error message gives the attacker. It also reveals the particular lines in the code where database operations take place as well as the internal directory structure of the server (/home/doau/public_html/...). By comprehending the file layout, authentication behavior, and backend configuration, these leaks together aid the attacker in improving their exploitation attempts.

2. (15 points) The architecture components Linux, Apache, MySQL, and PHP (a.k.a. LAMP) are often found installed on the same physical server. Why can this diminish the security posture of the application's architecture?

The security posture deteriorates when all LAMP components Linux, Apache, MySQL, and PHP run on the same physical server since a breach of any one layer can swiftly spread to the others. An attacker can directly access the system hosting the database (MySQL) and do complete data extraction or alteration if they take advantage of a weakness in the web application (PHP) or web server (Apache). Because of this lack of separation, there is also no defense-in-depth, no containment of attacks, and no isolation of privileges one breach effectively becomes a system wide breach. The impact of misconfigurations, resource depletion, and privilege escalation grows when all components are centralized on a single system, increasing the vulnerability of the entire stack.